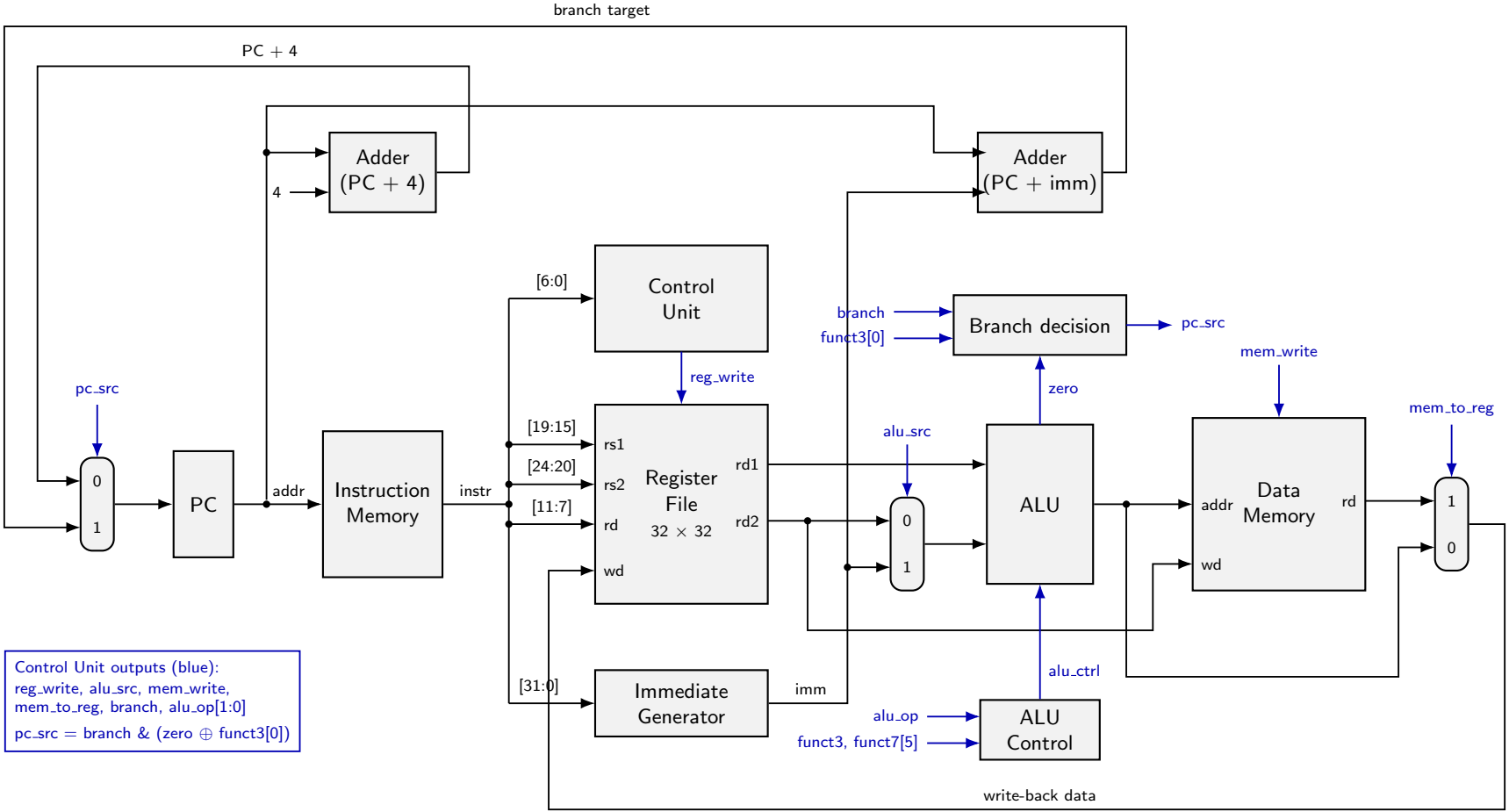
branch target
PC + 4
Adder
(PC + 4)
4
Adder
(PC + imm)
[6:0]
Control
Unit
branch
funct3[0]
Branch decision
pc_src
reg_write
zero
mem_write
mem_to_reg
pc_src
alu_src
[19:15]
[24:20]
[11:7]
rs1
rs2
rd
wd
Register
File
32 × 32
rd1
rd2
0
1
ALU
addr
Data
Memory
rd
wd
1
0
0
1
PC
addr
Instruction
Memory
instr
[31:0]
Immediate
Generator
imm
alu_ctrl
alu_op
funct3, funct7[5]
ALU
Control
write-back data
Control Unit outputs (blue):
reg_write, alu_src, mem_write,
mem_to_reg, branch, alu_op[1:0]
pc_src = branch & (zero ⊕ funct3[0])